

ABC456

A - Dice

分值: 100 分

Problem Statement

There are three dice, each with six faces.
Each die has 1, 2, 3, 4, 5, 6 written on its faces.

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Each die has 1, 2, 3, 4, 5, 6 written on its faces.

When these dice are rolled simultaneously, is it possible for the sum of the rolled values to be X ?

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Constraints

- X is an integer between 1 and 20, inclusive.
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-

Input

The input is given from Standard Input in the following format:

输入从标准输入按如下格式给出:

X

Output

If it is possible for the sum of the rolled values to be X , output Yes; otherwise, output No.

If it is possible for the sum of the rolled values to be X , output Yes; otherwise, output No.

Sample Input 1

15

Sample Output 1

Yes

For example, if the rolled values are 4, 5, 6, the sum is 15.

For example, if the rolled values are 4, 5, 6, the sum is 15.

Sample Input 2

2

Sample Output 2

No

B - 456

分值: 200 分

Problem Statement

There are three dice, each with six faces.

The j -th face of the i -th die has $A_{i,j}$ written on it.

For each die, each face comes up with probability $\frac{1}{6}$.

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The j -th face of the i -th die has $A_{i,j}$ written on it.

For each die, each face comes up with probability $\frac{1}{6}$.

When these dice are rolled simultaneously, find the probability that the values 4, 5, 6 each appear exactly once.

When these dice are rolled simultaneously, find the probability that the values 4, 5, 6 each appear exactly once.

Constraints

- $A_{i,j}$ is an integer between 1 and 6, inclusive.
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-

Input

The input is given from Standard Input in the following format:

输入从标准输入按如下格式给出:

```
A1,1 A1,2 A1,3 A1,4 A1,5 A1,6  
A2,1 A2,2 A2,3 A2,4 A2,5 A2,6  
A3,1 A3,2 A3,3 A3,4 A3,5 A3,6
```

Output

Output the answer. The answer is considered correct if the absolute or relative error from the true answer is at most 10^{-6} .

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Sample Input 1

```
1 2 3 4 5 6
1 2 3 4 5 6
1 2 3 4 5 6
```

Sample Output 1

```
0.0277777778
```

The desired probability is $\frac{1}{36}$. An output such as 0.027778 is also accepted.

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Sample Input 2

```
4 5 6 4 5 6
4 4 5 5 6 6
6 5 4 4 5 6
```

Sample Output 2

```
0.2222222222
```

The desired probability is $\frac{2}{9}$.

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